

Mattia Martini

Postdoctoral researcher

Contact info

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Experience

Postdoctoral researcher

Sept 2025 – now

Centre de Mathématiques Appliquées (CMAP), École polytechnique

- Mentors: Mao Fabrice Djete and Huyên Pham
- Part of the ANR PEPR project MIRTE

Postdoctoral researcher

Apr 2023 – Aug 2025

Laboratoire J.A. Dieudonné (LJAD), Université Côte d'Azur

- Mentor: François Delarue
- Member of the ERC project ELISA

Education

PhD Mathematical sciences, Università degli studi di Milano

Oct 2019 – Mar 2023

- Thesis: *Advances in stochastic analysis on spaces of measures: Kolmogorov equations related to stochastic filtering and mean field optimal stopping*
- Supervisor: Marco Fuhrman

MSc Mathematical engineering, Politecnico di Milano (110/110 cum laude)

Sept 2017 – Oct 2019

- Thesis: *Kolmogorov Equations on Space of Measures Associated to Stochastic Filtering Processes*
- Supervisors: Sandro Salsa and Marco Fuhrman

BSc Mathematical engineering, Politecnico di Milano (110/110)

Sept 2014 – Sept 2017

- Thesis: *Quantum bits and superdense coding*
- Supervisor: Matteo Gregoratti

High school diploma, I.T.I.S Galileo Galilei, (100/100 cum laude)

2009 – 2014

Publications

Preprints

- [1] F. Delarue, M. M., and G. E. Sodini. "HJB equations driven by the Dirichlet-Ferguson Laplacian in Wasserstein-Sobolev spaces". arXiv:2511.03522. 2025.
- [2] A. Cosso, M. M., and L. Perelli. "Mean convergence rates for Gaussian-smoothed Wasserstein distances and classical Wasserstein distances". arXiv:2504.17477. 2025.

Published papers

- [3] A. Cecchin, S. Daudin, J. Jackson, and M. M. "Quantitative convergence for mean field control with common noise and degenerate idiosyncratic noise". *Electron. J. Probab.* 30 (2025), pp. 1–62.
- [4] G. Franzese, M. M., G. Corallo, P. Papotti, and P. Michiardi. "Latent Abstractions in Generative Diffusion Models". *Entropy* 27.4 (2025).
- [5] F. Delarue and M. M. "Fourier Galerkin approximation of mean field control problems". *Nonlinear Differential Equations and Applications NoDEA* 32.1 (2024), p. 11.
- [6] M. M. "Kolmogorov equations on the space of probability measures associated to the nonlinear filtering equation: the viscosity approach". *Stochastic Analysis and Applications* 42.6 (2024), pp. 987–999.
- [7] A. Cosso and M. M. "On smooth approximations in the Wasserstein space". *Electronic Communications in Probability* 28 (2023), pp. 1–11.
- [8] B. Djehiche and M. M. "Time-inconsistent mean-field optimal stopping: A limit approach". *Journal of Mathematical Analysis and Applications* 528.1 (2023).
- [9] M. M. "Kolmogorov equations on spaces of measures associated to nonlinear filtering processes". *Stochastic Processes and their Applications* 161 (2023), pp. 385–423.
- [10] M. M. and G. E. Sodini. "Numerical Methods for a System of Coupled Cahn-Hilliard Equations". *Commun. Appl. Ind. Math.* 12 (2021), pp. 1–12.

Given talks, seminars and posters

Torino seminar series in Stochastics and Mathematical Statistics , Università di Torino	Apr 2026
Seminar of the probability and statistics group	
WS Mean-field systems in finance, neurosciences and AI , Folgarida, Trento	Jan 2026
Contributed talk	
Seminario di Probabilità e Statistica , Politecnico di Milano	Jan 2026
Seminar of the probability and statistics group	
Seminari in Probabilità e Finanza , Università di Padova	Dec 2025
Seminar of the probability and finance group	
Séminaire du Pôle Probabilités , École polytechnique (CMAP)	Oct 2025
Seminar of the probability group	
Interacting Particle Systems and Applications , Università di Siena	Sept 2025
Invited talk	
16th Viennese Conference Optimal Control and Dynamic Games , TU Wien	July 2025
In the invited session "Stochastic Control and Applications"	
Durham Symposium on Mean Field Games , Durham University	March 2025
Invited talk	
WS Optimal Transport: from robust pricing to model calibration , Folgarida, Trento	Feb 2025
Contributed talk	

Probability and Statistics Seminar , University of Luxembourg Seminar of the probability and statistics group	Jan 2025
PDE Seminar , Georgia Tech (online) Seminar of the PDE group	Sept 2024
2nd AMS-UMI International Joint Meeting , Università degli Studi di Palermo In the invited session "Mean Field Games and related topics"	July 2024
Seminario di Probabilità e Statistica , Politecnico di Milano Seminar of the probability and statistics group	June 2024
Fourth Italian Meeting on Probability and Mathematical Statistics , Roma In the invited session "Stochastic optimal control of McKean-Vlasov equations"	June 2024
Young Researchers in Stochastic Control and Games , Universität Bielefeld Invited talk	Dec 2023
Séminaire de Probabilités et Statistique , Université Côte d'Azur (LJAD) Seminar of the probability and statistics group	Dec 2023
PGMO Days , EDF Lab Paris-Saclay In the invited session "Mean field games and control"	Nov 2023
A Random Walk in the Land of Stochastic Analysis and Numerical Probability , CIRM Luminy Poster	Sept 2023
SPDEs, optimal control and mean field games , Universität Bielefeld Poster	July 2023
The 9th International Colloquium on BSDEs and Mean Field Systems , IAE Savoie Mont Blanc Contributed talk	July 2022
Third Italian Meeting on Probability and Mathematical Statistics , Università di Bologna In the contributed session "New trends in stochastic control"	June 2022
100 years UMI and First UMI meeting of PhD students , Università di Padova Contributed talk	May 2022
Seminars cycle "Insalate di Matematica" , Università degli Studi Milano-Bicocca Seminar organized by the PhD students	Nov 2021
Trento-Wuppertal School on Stochastic Processes, Analysis and Semigroups , Università di Trento Contributed talk	Sept 2021

Teaching

Stochastic Calculus and Mathematical Finance , Università degli Studi di Torino Lecturer MSc Quantitative Finance and Insurance	2026
Stochastic differential equations , Politecnico di Milano Teaching assistant MSc Mathematical Engineering	2022
Stochastic differential equations , Politecnico di Milano Teaching assistant MSc Mathematical Engineering	2021
Basics of Statistics , Politecnico di Milano Teaching assistant BSc Biomedical Engineering	2021
Stochastic differential equations , Politecnico di Milano Teaching assistant MSc Mathematical Engineering	2020

Visiting periods

Università degli Studi di Torino , 1 Month Visiting professor position	Apr-May 2026
University of Luxembourg , 1 Week Collaboration with Michele Stecconi	Jan 2025
Imperial College, London , 1 Week Collaboration with Ofelia Bonesini	Jul 2024
Università di Parma, Italy , 1 Week Collaboration with Alessandro Calvia	Feb 2024
KTH, Stockholm , 1 Week Collaboration with Boualem Djehiche	May 2022

Organization

Contributed session "(Stochastic) Analysis on Spaces of Probability Measures and Applications", with Giacomo Sodini

5th Italian Meeting on Probability and Mathematical Statistics, 2026 | Palermo

Workshop on Stochastic Processes, Stochastic Optimal Control, and their Applications, among the organizers
September 26 - 27th, 2024 | Politecnico di Milano

Contributed session "Mean Field Games and Mean Field Control I", with Jodi Dianetti

4th Italian Meeting on Probability and Mathematical Statistics, 2024 | Roma

Phd Days 2021, among the organizers

December 14 - 17th, 2021 | Università degli Studi di Milano

Others

Abilitations: Maître de conférences (26 - Mathématiques appliquées et applications des mathématiques)

Referee activities:

- Annals of Applied Probability, SICON, Stochastic Processes and Their Applications, Journal of Differential Equations, Systems & Control Letters, Statistics & Probability Letters
- ANR (French grant)

Participation to research projects:

- Sept 2025 - now: part of the ANR PEPR project MIRTE
- Apr 2023 - Aug 2025: member of the ERC ADV project ELISA (Grant agreement No. 101054746), P.I. F. Delarue.

Memberships:

- From 2025: Gruppo UMI Prisma (PProbability In Statistics, Mathematics and Applications)
- From 2022: Unione Matematica Italiana (UMI)
- From 2020: Indam, GNAMPA

Languages: Italian (Native) | English (Full professional proficiency) | French (Intermediate)